Supporting Information (appendix):

Sep & Vellinga, Sarabdjitsingh, Joëls (2021). *The rodent object-in-context task: a systematic review and meta-analysis of important variables.* PLOS ONE.

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A1. Complete PubMed Search string

(
("Memory, Episodic"[MeSH] OR "Memory, Long-Term"[MeSH] OR "Memory, Short-Term"[MeSH] OR "Spatial Memory"[MeSH] OR "Discrimination Learning"[MeSH] OR "Spatial Learning"[MeSH] OR "Episodic memory"[all Fields] OR "Memories"[all fields] OR "Episodic like memory"[all Fields] OR "long term memory"[all Fields] OR "short term memory"[all Fields] OR "spatial memory"[all Fields] OR "working memory"[all Fields] OR "discrimination learning"[all Fields] OR "spatial learning"[all Fields] OR "memory"[all Fields] OR "recognition memory"[all Fields] OR "Association learning"[MeSH] OR "Discrimination Learning"[MeSH] OR "Learning"[MeSH] OR "Learn"[all fields] OR "Learning"[all fields] OR "Learning task"[all Fields] OR "association learning"[all Fields] OR "discrimination learning"[all Fields] OR "discrimination task"[all fields])
AND
("rodentia"[MeSH] OR mice[all fields] OR mus[all fields] OR mouse[all fields] OR murine[all fields] OR woodmouse[all fields] OR rats[all fields] OR rat[all fields] OR murinae[all fields] OR muridae[all fields] OR cottonrat[all fields] OR cottonrats[all fields] OR hamster[all fields] OR hamsters[all fields] OR cricetinae[all fields] OR rodentia[all fields] OR rodent[all fields] OR rodents[all fields])
AND
(
"OIC"[all fields] OR "object in context"[all fields] "object context"[all fields] OR "object place context"[all fields] OR "object background"[all fields] OR "context mismatch"[all fields] OR "object-in-context"[all fields] OR "context item"[all fields] OR "context recognition"[all fields]
OR
(("object"[all fields] OR "item"[all fields]) AND "context"[all fields] AND ("task"[all fields] OR "test"[all fields] OR "recognition"[all fields] OR "discrimination"[all fields]))
)
)
NOT ("meta-analys*"[ti] OR "review"[ti] OR Review [Publication Type] OR systematic[sb] OR "meta-analysis"[Publication Type])

A2. Data-extraction codebook

Table A1. Data-extraction codebook
** Empty cells indicate that information was not (re)coded but extracted as described in the article.*
# Variables with more than 1/3 missing values that were not included in the random forest analysis.

Variable	General explanation	Coding*
Publication details		
Reference	APA reference (copied from google scholar)	
PMID	Pubmed ID	
Authors	From Pubmed	
Affiliation	From Pubmed	
year	Of publication	
journal	From Pubmed	
EXP_group	Identifier of each unique animal group in a study (unique in combination with PMID).	
Animal factors		
Species	Animal species. Note, although all rodent species were eligible, only rat and mice papers were present in the dataset.	Rat / Mice
Strain	Strain as reported in publication	
Sex	Sex of animals	M (Male) / F (Female) / Both
Origin	Origin of the breeding animals	B (in-house breeding) / P (purchased animals)
Age	Grouped age	Juvenile (0-21 days); Adolescent (22-56 days); Adult (57-365 days); Senior (>365 days).

Prev.Exp	Participation in previous experiments other than object recognition? (see comment for type of experiment)	Yes / No
Prev.OR	previous participation in object recognition experiments	Yes / No
Type.Control	Type of control group	U (Untouched) / SO (sham operated) / SI (saline injected)
Experimental factors		
AdLib	Ad libitum access to food or water?	Yes / No
Housing	Housing prior to experiment	G (group-housed) / S (single-housed)
RevRhythm	Reversed day-night rhythm?	Yes / No
nC	Sample size of control group; Note,the mean size is calculated if only a range is provided.	
Task-related experimental factors		
Context.Floor	Experimental box; contextual differences; floors box A/B?	Different / Similar
Context.Wall	Experimental box; contextual differences walls box A/B?	Different / Similar
Context.Room	Experimental box; contextual differences; rooms box A/B?	Different / Similar
Context.Size.A	Floor size of box A	Size (cm2)
Context.Size.B	Floor size of box B	Size (cm2)
Context.Shape	Experimental box; shape difference between box A/B?	Different / Similar
Object.Material	Objects; characteristics; different materials?	Different / Similar
Object.Size #	Objects; characteristics; different sizes?	Different / Similar
Object.Position #	Objects; Position objects relative to room / each other	
Animal.position #	Position how the animal is placed in the box.	
Handling #	Handling prior experiments	Yes / No
Testroom.Habituation #	Habituation to test room	Yes / No
Transport.Habituation #	Habituation to transportation	Yes / No
Context.Habituation.Freq	Habituation to experimental boxes; Number of visits per box	
Habituation.Time.Trial	Habituation to experimental boxes; time per trial	Time (min)
Habituation.Time.Total.Context	Total habituation time per box	Time (min)
Habituation.with.Cagemates	Habituation with cage mates?	Yes / No / Partly
Object.Habituation	Habituation to experimental boxes; objects present?	Yes / No / Partly
Learning.Time.Trial	Learning (acquisition) phase per trial	Time (min)
Learning.Learning.Delay	time between learning trials	Time (min)
Learning.Order.Counterbalanced	Learning sequence counterbalanced?	Yes / No
Learning.Exploration.Time.A #	Learning phase; average exploration time box A	Time (sec)
Learning.Exploration.Time.B #	Learning phase; average exploration time box B	Time (sec)

Learning.Exploration.Time.Total #	Sum of average exploration time in box A and B	Time (sec)
Learning.Memory.Delay	Retention time between last learning trial and the memory trial	Time (min)
Memory.Time.Trial	Memory phase; time of trial	Time (min)
Memory.Exploration.Time.A #	Memory trial; exploration time box A	Time (sec)
Memory.Exploration.Time.B #	Memory trial; exploration time box B	Time (sec)
Memory.Exploration.Time.Familiar #	Memory trial; exploration time in context object	Time (sec)
Memory.Exploration.Time.New #	Memory trial; exploration time out of context object	Time (sec)
Memory.Exploration.Time.Total #	Total exploration time in memory trial (average of expl box A and B or sum in and out of context objects)	Time (sec)
Context.Order	Context order: AA (Last box learning similar to memory box), AB (Last box learning different from memory box), Both (randomized for context order). Note, the counterbalanced use of AB and BA is coded as AB; Both us used for AA and AB combinations.]	AA/AB/Both
Scoring	Behavioral Scoring	Manual/Digital/Both
Program	Behavioral scoring, if digital: Name of scoring program + version	
Observers #	Behavioral scoring, if manual: Number of observers	1; >1
Def_Expl	Definition of exploration	
SmellCtrl	“Smell control”: cleaning box / object or changing bedding)	Yes / No
ReuseBedding #	Reuse of OIC bedding?	Yes / No (if yes, number of trials added)
Scoring		
DR.Formula	Formula for DI/DR	
Page_outcome	Location (Page number) of outcome (DR) in article	
Memory.Type	Memory domain, based on retention time between learning and memory phase: Short-term memory (STM), ‘medium, long-term memory (LTM)	STM (0-10min delay) / ‘medium’ (10 min. -1hr delay) / LTM (>1hr delay)
DR	Discrimination ratio, as reported in article or extracted from figures with PlotDigitizer	
SEM	Standard error of the mean from DR	
DR_B	(Converted) DR_B_n/(n+f); DR_A *0.5+0.5	Range: 0 -1
DR_A	(Converted) DR_A_=(n-f)/(n+f): DR_B *2-1	Range: -1 -1
SEM_DR_B	(Converted) SEM corresponding to DR_B: SEM_A / 2	
SEM_DR_A	(Converted) SEM corresponding to DR_A: SEM_B * 2	

A3. DR conversion formula's

A conversion formula was developed to convert the DR that results from one formula into the other. The original two different DR formulas are as follows:

$$\text{DR_A} = \frac{\text{Tnovel} - \text{Tfamiliar}}{\text{Tnovel} + \text{Tfamiliar}} \quad \text{or} \quad \text{DR_B} = \frac{\text{Tnovel}}{\text{Tnovel} + \text{Tfamiliar}}$$

DR_A calculates the difference between the exploration time of both objects (T_{novel}-$T_{familiar}$) divided by the total amount of object exploration (T_{total}) during the test phase. DR_A ranges from -1 to 1, making 0 the score at which both objects are explored equally. DR_B calculates the fraction of exploration of the novel object by dividing T_{novel} by T_{total}. DR_B has a range from 0 to 1, this makes 0.5 the score at which animals explore both objects for an equal amount of time.

All DRs that originated from DR_B were converted to scores comparable with DR_A.

To calculate the conversion-factor the formulas were equated to each other. A multiplication factor (*x*) was added to DR_B to correct for the broader range of DR_A (-1 to 1) compared to DR_B (0 to 1). Besides a value (*y*) was added to correct for the shift of the score at which both objects are explored for an equal amount of time (DR_A = 0 and DR_B = 0.5). This results in the following equation:

$$\frac{Tnovel - Tfamiliar}{Tnovel + Tfamiliar} = \frac{Tnovel}{Tnovel + Tfamiliar}x + y$$

To determine the values of *x* and y, two points (T_{novel}, $T_{familiar}$) were inserted in the equation. These points were point 1 (*a,b*) and point 2 (*c,d*). This gives;

1. $(a-b)/(a+b) = a/(a+b)\,x + y$ **and**
2. $(c-d)/(c+d) = c/(c+d)\,x + y \rightarrow y = (c-d)/(c+d) - c/(c+d)\,x$

↓Insert this *y* value into equation 1, this yields;

$$(a-b)/(a+b) = a/(a+b)\,x + (c-d)/(c+d) - c/(c+d)\,x$$
$$(a/(a+b) - c/(c+d))x = (a-b)/(a+b) - (c-d)/(c+d)$$
$$\big(a(c+d) - c(a+b)\big)x = (a-b)(c+d) - (a+b)(c-d)$$
$$(ac + ad - ac - bc)x = ac + ad - bc - bd - ac + ad - bc + bd$$
$$(ad - bc)x = 2ad - 2bc$$
$$x = (2ad - 2bc)/(ad - bc)$$
$$x = 2$$

↓Insert *x*=2 back into the *y*-equation that resulted from equation 2, this gives;

$$y = (c-d)/(c+d) - c/(c+d) * 2$$
$$y = (c - d - 2c)/(c+d)$$
$$y = (-c - d)/(c+d)$$
$$y = -1$$

Now the *x* and *y* values are known they can be inserted in the conversion formula;
$\frac{Tnovel - Tfamiliar}{Tnovel + Tfamiliar} = 2 * \frac{Tnovel}{Tnovel + Tfamiliar} - 1 \rightarrow$ This way DR_B scores can be converted by multiplying the original score by 2 and subtracting 1; $DR_A = DR_B * 2 - 1$.

Final conversion formula DR: $\mathbf{DR_A = 2 * DR_B - 1}$

As the range of DR_A is twice as large as the range of DR_B, the SEM of DR_B is multiplied by 2 to create the SEM of DR_A.

Final conversion formula SEM: SEM DR_A = SEM DR_B * 2

A4. Summary quality scores formulas

Randomization score = Sequence generation (yes = 1, no/unclear = 0) + Baseline characteristics (yes = 1, no/unclear = 0) + Allocation concealment (yes = 1, no/unclear = 0) + Random housing (yes = 1, no/unclear = 0) + Random outcome assessment (yes = 1, no/unclear = 0)

Blinding score = Blinding intervention (yes = 1, no/unclear = 0) + Blinding (analysis) (yes = 1, no/unclear = 0)

Reporting score = Incomplete outcome data (yes = 1, no/unclear = 0) + Selective outcome reporting (yes = 1, no/unclear = 0)

Summary quality score = Randomization score + Blinding score + Reporting score

A5. Summary score formulas

Context difference score (Context.Difference.Score) = Floor (Different = 4, Similar = 0) + Walls (Different = 3, Similar = 0) + Shape (Different = 1, Similar = 0) + Rooms (Different = 1, Similar = 0).

Arousal prior to OIC (Arousal.Prior) = Animal origin (purchased = 1, in-house breeding = 0) + Previous participation in experiments (Yes = 1, No = 0) + Previous participation in Object Recognition tasks (Yes = 0, No = 1) + Type of control group (Sham operation = 2, Sham injection = 1, Undisturbed = 0) + Ad libitum food access (Yes = 0, No = 1) + Reversed Day/night cycle (Yes = 0, No =1) + Housing (Group = 0, Single = 1).

Arousal related to OIC habituation procedures (Arousal.Task.Habituation) = Habituation to experimental boxes (Yes = 0, No = 1) + Habituation to objects (Yes = 0, Partly = 0, No = 1, n.a. = 0) + Habituation with cage mates (Yes = 0, Partly = 0, No = 1, n.a. = 0). Note, n.a. (not applicable) was assigned to studies without habituation phase.

Arousal.Total = Arousal prior to OIC + Arousal related to OIC habituation procedures

A6. Study characteristics

Table A2 Study characteristics of included studies
not included in meta-analysis, only included in systematic review

Study	Species	Sample size
(Ameen-Ali, Eacott, and Easton 2012)	Rats	5
(Asiminas et al. 2019)	Rats	12, 12, 12
(Balderas et al. 2008)	Rats	7, 5, 7, 7, 6, 6, 8, 10
(Barsegyan, McGaugh, and Roozendaal 2014)	Rats	13,12
(Bekinschtein et al. 2013)	Rats	8.5
(Bukhari, Clark, and Williamson 2018)	Rats	5.25, 5.25, 7.25, 7.25
(Chomiak et al. 2014)	Rats	5
(Czerniawski et al. 2015)	Rats	12
(Dix and Aggleton 1999)	Rats	14
(Heimer-McGinn et al. 2017)	Rats	6
(Kanatsou et al. 2015)	Mice	14
(La Spina et al. 2019)	Rats	7, 7
(Langston and Wood 2010)	Rats	13
(Lee et al. 2014) #	Rats	10

(Ji-Tao Li et al. 2013)	Rats	10
(Ji-Tao Li et al. 2011)	Rats	11, 12
(J-T Li et al. 2015)	Rats	8, 9
(Liu et al. 2018)	Rats	11, 11
(Lomeli et al. 2017)	Rats	25
(MacIlvane et al. 2016)	Rats	8, 8
(Martínez et al. 2014) #	Rats	
(Morici et al. 2018)	Mice	12, 13, 13.5, 13.5, 14, 14
(Mumby 2002)	Rats	14
(Mychasiuk et al. 2015)	Rats	3, 3
(Noble et al. 2019)	Rats	11, 12
(Norman and Eacott 2005)	Rats	11, 11
(Pillai et al. 2018)	Mice	6
(Ramsaran, Sanders, and Stanton 2016)	Rats	9, 13, 15, 10, 10, 13, 13, 10, 11
(Ramsaran, Westbrook, and Stanton 2016)	Rats	16, 21, 24, 20
(Sasaki Russell et al. 2019) #	Rats	12
(Sanders, Heroux, and Stanton 2020)	Rats	22.75, 22.75
(Sanderson et al. 2011)	Mice	8, 16
(Schulz et al. 2011)	Rats	9, 9
(Sigwald et al. 2019)	Rats	6
(Spanswick and Dyck 2012)	Mice	5
(Spanswick and Sutherland 2010)	Rats	10
(Tam, Bonardi, and Robinson 2015)	Rats	6, 15, 6, 15
(Vandrey et al. 2020) #	Mice	17
(Wilson et al. 2013)	Rats	7, 8
(Woodward et al. 2018)	Rats	16
(Zhao et al. 2007)	Mice	10.5

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A7. Systematic review table

Table A3. Systematic overview of Variation in the OIC

Characteristic	Observations (N = 97[1])
Sample size	3.0 - 25.0; 10.8; 10.9 (4.6)
Not reported	13
Species	
Mice	8 / 97 (8.2%)
Rat	89 / 97 (92%)
Strain	
C57BL/6	3 / 94 (3.2%)
Dark Agouti rats	2 / 94 (2.1%)
ICR	1 / 94 (1.1%)
Lister hooded rats	8 / 94 (8.5%)
Long-Evans rats	22 / 94 (23%)
pigmented DA strain	1 / 94 (1.1%)
Sprague-Dawley	26 / 94 (28%)
Sv/Ev mice	1 / 94 (1.1%)
Tg(Sim1cre)KH21Gsat/Mmucd	1 / 94 (1.1%)
Wistar rats	29 / 94 (31%)
Not reported	3
Sex	
Female	6 / 95 (6.3%)
Male	67 / 95 (71%)
Mixed	22 / 95 (23%)
Not reported	2
Origin	
In-house breeding	46 / 74 (62%)
Supplier	28 / 74 (38%)
Not reported	23
Age	
Adolescent	11 / 90 (12%)
Adult	67 / 90 (74%)
Juvenile	10 / 90 (11%)
Senior	2 / 90 (2.2%)
Not reported	7
Type.Control	
Undisturbed	36 / 97 (37%)
Sham injection	38 / 97 (39%)
Sham surgery	23 / 97 (24%)

Table A3. Systematic overview of Variation in the OIC

Characteristic	Observations (N = 97[1])
Housing	
Group	67 / 92 (73%)
Single	25 / 92 (27%)
Not reported	5
Reversed day/night cycle	
No	78 / 80 (98%)
Yes	2 / 80 (2.5%)
Not reported	17
Previous experiment	
No	79 / 95 (83%)
Yes	16 / 95 (17%)
Not reported	2
Previous object recognition task	
No	61 / 95 (64%)
Yes	34 / 95 (36%)
Not reported	2
Ad libitum food & water	
No	3 / 93 (3.2%)
Yes	90 / 93 (97%)
Not reported	4
Handling [#]	
Yes	52 / 52 (100%)
Not reported	45
Context.Floor	
Different	68 / 80 (85%)
Similar	12 / 80 (15%)
Not reported	17
Context.Wall	
Different	88 / 95 (93%)
Similar	7 / 95 (7.4%)
Not reported	2
Context.Shape	
Different	37 / 96 (39%)
Similar	59 / 96 (61%)
Not reported	1
Context.Room	
Different	25 / 56 (45%)
Similar	31 / 56 (55%)

Table A3. Systematic overview of Variation in the OIC

Characteristic	Observations (N = 97[1])
Not reported	41
Context.Size.A	360 - 10,000; 2,400; 3,051 (1,823)
Not reported	3
Context.Size.B	289 - 10,000; 2,400; 3,101 (1,835)
Not reported	3
Object.Material	
Different	60 / 78 (77%)
Similar	18 / 78 (23%)
Not reported	19
Object.Size [#]	
Different	33 / 62 (53%)
Similar	29 / 62 (47%)
Not reported	35
Context.Order	
AA	32 / 95 (34%)
AB	23 / 95 (24%)
Both	40 / 95 (42%)
Not reported	2
Transport.Habituation [#]	
Yes	18 / 18 (100%)
Not reported	79
Testroom.Habituation [#]	
Yes	22 / 22 (100%)
Not reported	75
Habituation.with.Cagemates	
n.a.	19 / 72 (26%)
No	43 / 72 (60%)
Partly	8 / 72 (11%)
Yes	2 / 72 (2.8%)
Not reported	25
Context.Habituation.Freq	0.00 - 8.00; 3.00; 2.91 (1.89)
Not reported	8
Habituation Time Trial (min)	0 - 20; 10; 9 (6)
Not reported	17
Habituation Time Total Context (min)	0 - 120; 30; 27 (19)
Not reported	9
Object.Habituation	
n.a.	7 / 88 (8.0%)

Table A3. Systematic overview of Variation in the OIC

Characteristic	Observations (N = 97[1])
No	67 / 88 (76%)
Partly	10 / 88 (11%)
Yes	4 / 88 (4.5%)
Not reported	9
Learning Time Trial (min)	1.8 - 20.0; 5.0; 7.2 (4.4)
Learning-Learning Delay (min)	0 - 1,440; 5; 232 (493)
Not reported	4
Learning.Order.Counterbalanced	
No	3 / 83 (3.6%)
Yes	80 / 83 (96%)
Not reported	14
Learning Exploration Time A (sec) [#]	15 - 158; 56; 59 (38)
Not reported	56
Learning Exploration Time B (sec) [#]	13 - 160; 56; 56 (39)
Not reported	56
Learning Exploration Total (sec) [#]	27 - 312; 111; 113 (75)
Not reported	54
Learning-Memory Delay (min)	1 - 5,760; 120; 743 (876)
Not reported	2
Memory Time Trial (min)	0.50 - 10.00; 3.00; 3.26 (1.54)
Not reported	3
Memory Exploration Time A (sec) [#]	24 - 55; 37; 38 (15)
Not reported	93
Memory Exploration Time B (sec) [#]	19 - 58; 45; 42 (18)
Not reported	93
Memory Exploration Time Familiar (sec) [#]	8 - 37; 13; 16 (9)
Not reported	87
Memory Exploration Time New (sec) [#]	11 - 51; 22; 25 (13)
Not reported	87
Memory Exploration Time Total (sec) [#]	11 - 69; 25; 32 (17)
Not reported	64
Smell Control OIC	
Yes	75 / 75 (100%)
Not reported	22
Reuse Bedding OIC [#]	
n.a.	43 / 55 (78%)
No	1 / 55 (1.8%)
Partly	1 / 55 (1.8%)

Table A3. Systematic overview of Variation in the OIC

Characteristic	Observations (N = 97[1])
Yes	10 / 55 (18%)
Not reported	42
Scoring	
Both	2 / 73 (2.7%)
Digital	11 / 73 (15%)
Manual	60 / 73 (82%)
Not reported	24
Observers [#]	
>1	23 / 54 (43%)
1	20 / 54 (37%)
n.a.	11 / 54 (20%)
Not reported	43
Scoring Program	
AnyMaze software	8 / 71 (11%)
Ethovision XT (Noldus)	2 / 71 (2.8%)
Limelight2 program (Actimetrics; Coubourn Instruments)	1 / 71 (1.4%)
Med-PCIV software	1 / 71 (1.4%)
n.a.	57 / 71 (80%)
Topscan	2 / 71 (2.8%)
Not reported	26
DR.Formula	
(tnew-tfamiliar)/(tnew+tfamiliar)	36 / 97 (37%)
(tnew)/(tnew+tfamiliar)	39 / 97 (40%)
[(tnew-tfamiliar)/(tnew+tfamiliar)]*100	3 / 97 (3.1%)
[(tnew)/(tnew+tfamiliar)]*100	19 / 97 (20%)

[1]Range; Median; Mean (SD); n / N (%); [#] Variables with more than 1/3 missing values that were not included in the random forest analysis.

A8. Study quality

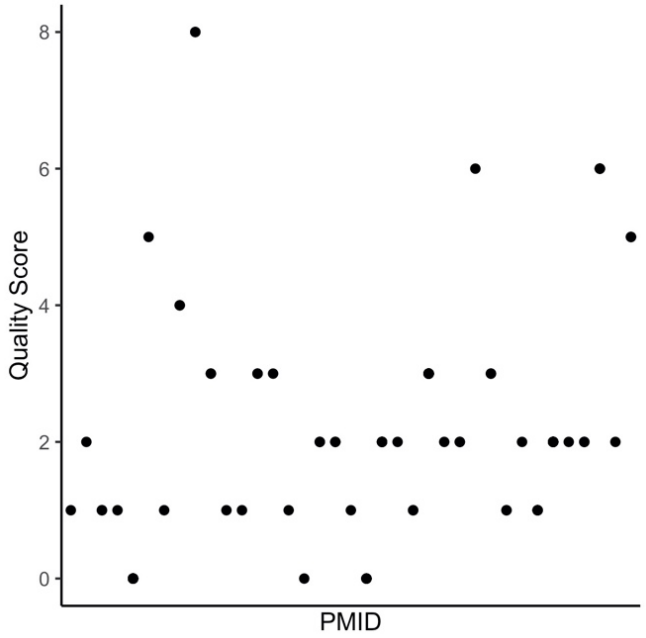

Fig A1. Cumulative study quality based on SYRCLE's risk of bias assessment per article.

A9. Random forest model convergence plot

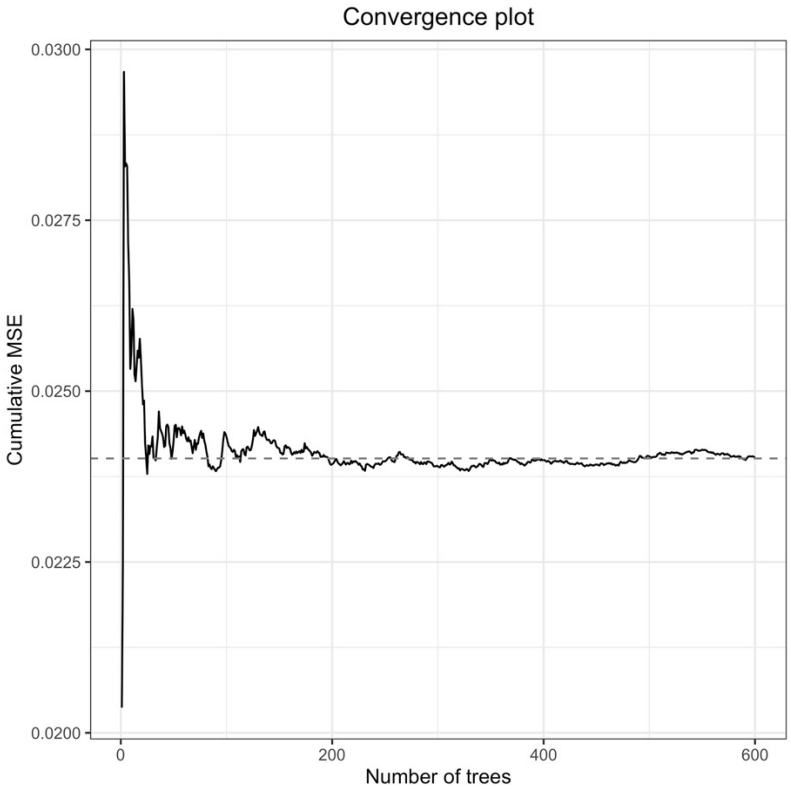

Fig A2. Random forest model convergence.
The stable cumulative MSE indicates good model convergence at 600 trees.

A10. PRISMA 2009 Checklist

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3-5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5; Appendix A1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6-7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Appendix A2
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7-8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7-8

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7-8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	8-9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Appendix A6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Fig 3
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Fig 4
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	10
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	11; Fig 5; Appendix A8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	11-12
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14-19
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15-16
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	19-21
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	22